ceotherm®

INDUSTRIAL FURNACES



INDUSTRIAL FURNACES AND EQUIPMENT

www.ceotherm.com

Ceotherm produces industrial furnaces, machinery and auxiliary equipment for aluminum melting and holding furnaces, heat treatment furnaces and machineries for aluminum extrusion, aluminum melting & recycling and aluminum forming industries.

Ceotherm also has products for ferrous and non-ferrous industries. Tailor-made design, manufacturing and application with modern Technologies and knowledge are performed by Ceotherm experience, engineering and technical capability.



Foundation Year	September 2010
Products and	Industrial Furnaces, design, manufacturing include electric, mechanic, burning system with related auxiliary equipment and control system
Number of Employees	>80 50-Blue collars for manufacturing, 12 technicians and supervisors 12 Engineers 6 finance and management
Production Area	3000 m²
Storage Area	500 m² + contacted logistic warehouses
Certificates	• ISO 9001:2014 Quality Management, • CE Quality Certificate • Certificate of Conformity for Individual products

PRODUCTS & SERVICE

Aluminum Recyling Furnace and Equipment For Billet and Sheet Ingots Production



- Reverberatory Type Melting Furnaces
- Holding and Casting furnaces
- Sidewell furnaces
- Double chamber furnaces
- Stationary type furnaces
- Dry Hearth
- Mechanical (vortex) pump
- Electromagnetic stirring system
- Regenerative burners System
- Scrap Charging Machine
- Liquid Metal Transfer Launders
- Rood Feeder Machines
- Degassing Units
- Ceramic Filter Units
- Aluminum Billet Homogenization Plant
- Automatic Billet Cutting Saw
- Automatic Billet Stacker/Destacker System
- Automatic Marking – Weighing



Aluminum Recyling Furnace and Equipment for ingot production



- Tilting Rotary Furnaces
- Ingot Casting Conveyors
- Jet Pulse Filter System

Furnaces For Extrusion Plant



- Billet Heating Furnaces
- Die Heating Furnaces
- Ageing Furnaces

Special Purpose Heat Treatment Furnaces

- T4-T6 Heat Treatment Furnace
- Foil & Coil Annealling Furnace
- Slab & Ingot Heating Furnace

01 Tilting - Rotary Type Melting Furnaces

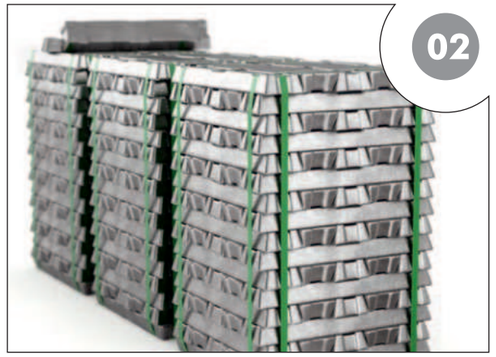
- Tilting rotary furnaces offer highly efficient recycling solutions for melting (recovering) aluminum scraps and dross obtained as residual material from other aluminum melting solutions.
- Especially oily, painted, and dirt materials, etc. Highly efficient in the recycling of aluminum scraps mixed with foreign materials is provided with tilting rotary furnaces.
- Suitable for melting slag, hard scrap, compressed scrap, painted and dirty scraps.
- Large production capacity from 500kg to 20ton
- Safe mechanical system with chain or direct coupled drive system on roller bearing
- Hydraulic tilting system
- Hydraulic controlled door system
- Max. Temperature:950°C
- Low energy consumption 750kW /ton
- Combustion system with oxy-fuel, fuel-oil or natural gas
- PLC controlled operating system with HMI interface
- Easy to use and reduced labor and minimum operating costs



02 Ingot Casting Machine

- Between 50-300 mould capacity
- Casting machine conveyor
- Ingot moulds and casting wheel
- Casting drive unit with speed control
- Tandish and mould pre- heating system
- Ingot knockout device
- Air and water cooling system
- Automatic stacking, stripping, weighing and marking system

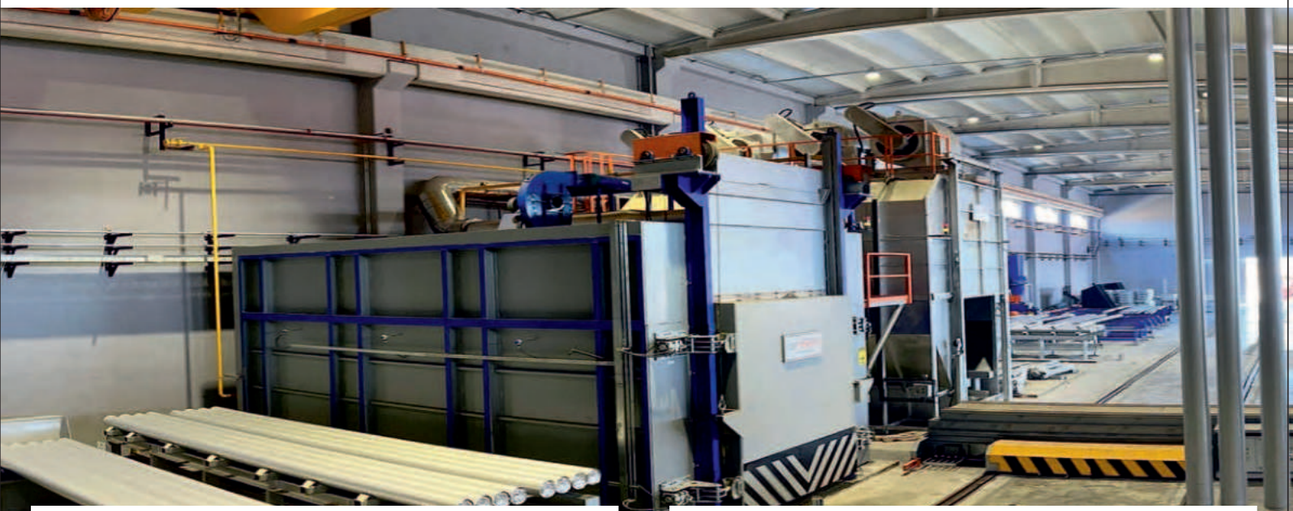




03 Jet Pulse Filter System

- Ambient air cleaning,
- As separation from flue gas in combustion processes,
- The highest efficiency (up to 99.99%) used in separating the conveyed (solid) and carrier (air) in pneumatic conveying systems It ensures that the dust and granules dispersed in the air in the production facilities are absorbed without spreading to atmostphere
- With best filtration efficiaency it helps maintain and quality
- It Contributes to the protection of occupational and worker health and increasing safety
- Easy maintenance and filtering element/bag replacement









04 Scrap Charging Machine

- It ensures that the aluminum ingots and scraps to be loaded into the aluminum melting furnace in the most optimum time without causing time and energy loss.
- System integration through on-network connection for applying the operation receipts
- Automatic and Manual operation
- Large display for visualization
- The charging car performs its right/left movement on the rails positioned in front of the furnace with hydro motors or reducers. During charging, the charging basket moves forward/backward with the hydraulic cylinder on the trolley, and it loads scrap or ingot directly to the furnace, even at high temperatures. 1 charging car easily serves 3 melting furnaces positioned side by side.



05 CeoMelt - Reverberatory Type Melting Furnaces

- 5 to100 Tons capacity
- Low energy consumtion with new technology regenerative burners Average gas consumption <60Nm^3 NG/ Ton Al.
- Direct connection to filtering system for to environmental protection
- Pneumatic cylinder-driven door clamping system.
- Low charging and skimming losses
- Air-cooled exhaust damper system.
- Rigit construction with robust steel body
- Easy handling and minimal losses with furnace geometry
- 85 %Al2O3 high alümina non-stick castable refractory No corundum built up.
- Hydraulic cylinder driven main door.
- Hydraulic tilting system.
- Stationary design with tap-out or transfer pump
- Electromagnetic stirring system
- Pre-degassing with porous plug system
- Modern operator interface and Industry 4.0
- PLC/HMI operated automation



06 Billet Casting System

- **Hydraulic cylinder**

Cylinder is mounted vertically in to the casting pit
Mounting flange at cylinder head

- **Lifting platform**

On the upper part of the lifting platform there are the connection holes for installation the dumpy billet seat.

- **Tilting device of casting platform**

It is composed of tipping frame, pedestal, turnover cylinder, position limitation device, water distribution device.

- **Support at well mouth**

Adjust height and support casting platform

- **Cooling water system**

This system is provided with water cooling tower and water pumps and the pipeline, manual butterfly valve and automatic cleaning filter. Air butterfly valve is controlled by control switch of the emergency water line, air regulating ball valve, pressure sensor, electromagnetic flow meter, temperature sensor, anti-vibration resilient joint, rotating joint.
Drainage system; A strong water drainage system is composed of one submerged pumps and liquid level transducer.

- **Hydraulic pipeline**

1Cr18Ni9Ti stainless steel pipes, hydraulic fittings with flat seals, hoses are high pressure rubber pipes with two layer steel wires



07 Degassing Unit & Ceramic Filter Unit

Degassing Unit

- In order to eliminate the problem that the hydrogen gas dissolved in liquid aluminum forming bubbles and gas pores during casting, inert gas «nitrogen (N) or argon (Ar)» injecting through rotating shafts and porous plugs to the molten metal and hydrogen gas is expelled. At the same time, it provides homogeneity the movement (mixing) of the molten metal during nitrogen or argon gas transitions.
- Nitrogen and argon gas supply line
- Reduction of 0.12-0.20 cc/100 g of safe hydrogen
- Process gas: Ar and *N*2
- Rotary gas injectors
- Electric preheating system with Sialon Tubes
- Easy heater and rotor replacement
- Fully automated PLC & HMI controlling

Ceramic Filter Unit

- Ceramic foam filters are used to filtering of molten aluminum.
- Preheating burner or electrical resistance installed on top cover.
- It is available in a range of size from in 7”-26” and grades 10 to 50 ppm.
- Refractory lined vessel is made from precast refractory.
- There is a drain for pouring remain aluminum after casting.

ALUMINUM RECYCLING FURNACES AND EQUIPMENT FOR BILLETS





08 AlTiB5 Rod Feeder

- In order to refine the grain structure of the liquid aluminum transferred from the melting furnace to the casting machine by transfer launders, AlTiB5 will be added into liquid aluminum by using an automatic controlled reducer driven Bar Feeding Machine for the AlTiB5 bar, which is served with coil rolls before the degassing unit.
- Double bar fed
- Adjustable Bar feeding speed 10-50 cm/minute
- Coil replacement alarm
- Coil outer diameter: 800 mm
- Coil width: 350 mm
- Coil Weight: 200 kg/coil
- Rod diameter approx. 9-10 mm



09 Billet Homogenizing Furnaces

- Best homogenizing quality to uniform for all billets
- 10-45 ton/charge capacity
- Working temperature : 580 ° C
- Improved temperature uniformity : ±5 ° C
- High efficency self recuperative or cold air burners
- Gas consumption<24Nm^3 Ng/Ton Al
- Reversible air circulation system
- Heating time: 8-11 hour depends on billet diameter
- E-motor reducer driven door
- Pneumatic actuated door clamping system
- Pneumatic actuated door locking system
- Explosion protection cover
- Automatic billet transfer car
- Cooling cabinet
- Reversible air flow cooling system
- Cooling rate up to 400 ° C/h
- Industry 4.0 PLC/HMI, easy operation and low labor costs





10 Aluminum Billet Head Cutting

- Between Ø90- Ø254 mm
- Cut length h tolerance +/- 5mm
- Cut angle tolerance 90 ° +/-0.5 °
- Automatic lubrication system
- The blade rotated by a motor
- Pneumatic billet clamping system
- Cutting performed with a single blade.
- The blade moved with a reducer
- Controlled blade speed by motor driver.
- Chip collection cyclone and suction fan
- Automatic chip collecting unit
- Blade thickness 6mm
- E-motor-gearbox driven input-output conveyor.
- Hydraulic throwing system
- Automatic billet weighing system
- Automatic marking for alloy code weight date and company logo to the billet surface font size between 8-40mm



11 Automatic Billet Stacker & Destacker system

- Stacking and de-stacking time <1 hours
- One common loading & unloading system can serve up to 4 furnaces
- E-motor reducer driven or hydraulic rope lifting system
- E-motor reducer driven movement system
- E-motor reducer driven point system
- Billets are automatically transferred to the cold saw machine one by one
- PLC & HMI control

12 Aluminum Ageing Ovens

Aluminum extrusion aging furnace/oven is mainly used for the final hardening processing of the finished product and quickly improves the intensity of aluminum profiles, which is conducive to the operation, storage, and transportation of subsequent processes, and is conducive to normal use.





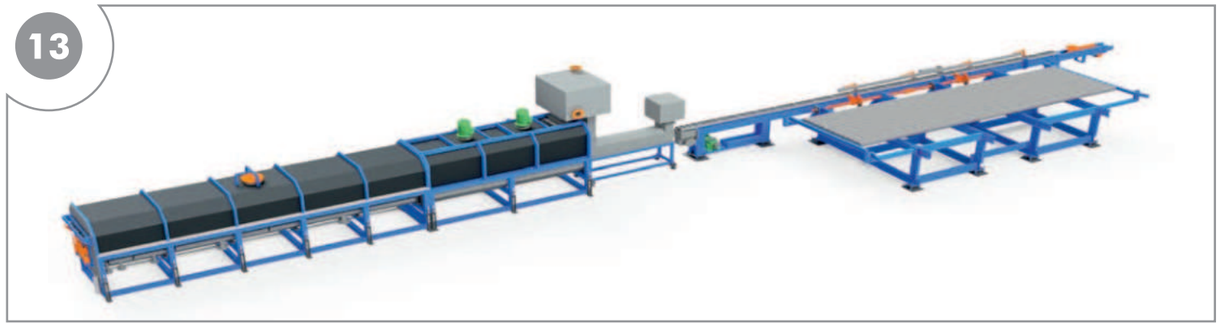
- The aging thermal cycle allows the production of the desired metallurgical properties such as hardness.
- Up to 30 ton/charge capacity
- Max. Temperature 250° C
- Working temperature 180-200C
- Temperature homogeneity ±3
- Treatment time 8 hours
- Indirect gas heating
- Low consumption 12-14 Nm^3 NG/ton AL production
- Furnace atmosphere circulating reversible flow
- Loading & Unloading by chain conveyor or shuttle car or roller conveyor options.





13 Aluminum Billet Heating Furnaces

- 500 kg/h to 5000 kg/h Capacity
- 2 to 4 Heating zones
- Working temperature 450-480 °C
- Natural gas heating system
- Natural gas consumption <20 Nm^3 NG / Ton Al
- Temperature homogeneity +/- 10 °C
- Billet Loading conveyor
- Hydraulic operated hot shear for zero waste to scrap



The billet furnace is typically used to preheat the aluminium billets to the desired temperature before loading them into the extrusion press.

The billet furnace has multiple zones for soaking, preheating, and heating. The soaking zone enables every aluminium billet to have a temperature uniformity of ±9 when it is fed out from the soaking zone.

The preheating zone is equipped by equipping with an additional air blower for efficient preheating performance, which exploits hot gases exhausted from the heating zone to impinge the aluminium billet that enters the pressure-controlled furnace chamber.

Unique nozzle design helps combustion to be optimal, enhancing temperature uniformity and thermal efficiency.

Due to its rapid heating speed, after being heated for 30 minutes as usual, the aluminium billet can be transferred into the extrusion process.

With features of high automation and flexibility, this single billet furnace is suitable for extruding a variety of industrial profiles.

Can be equipped with auxiliary equipment, such as a billet manipulator and hot-share billet cutter.

14 Die Heating Furnaces

- Independent heating chambers
- Max. Temperature 450-500 °C
- Temperature uniformity ±5 °C
- Heating system Electrically
- Heating element resistance
- Doors on top with pneumatic cylinder actuated
- Loading/unloading manually by crane





15 Stationary & Tilting Crucible Furnace

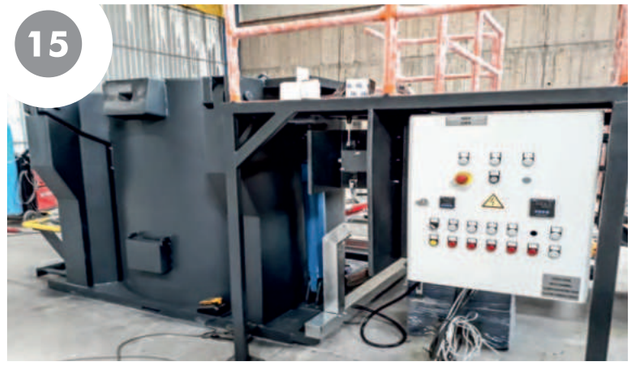
Gas Heated Furnaces

- Working temperature max. 1200° C
- Molten metal capacity 180-1000 kg
- Heating power 100-650 kW
- Power supply 380V- 50 Hz
- Melting capacity : 50-550 kg Al/h

Electrical Heated Furnaces

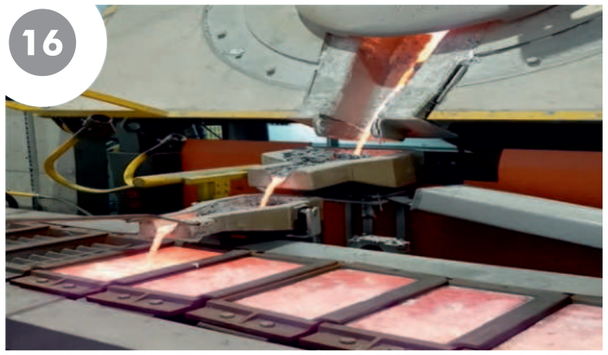
- Working temperature max. 1200° C
- Molten metal capacity 115-1150 kg
- Heating power 25-125 kW
- Power supply 380V- 50 Hz
- Melting capacity : 45-225 kg Al/h





16 Aluminum Melting Induction Furnaces

- For the best efficiency and mixing the molten metal.
- Low Frequency solid state
- Paralel resonance power coverters
- Converter power 50 kw -6MW
- Furnace capacity 100KG-10 TON
- Aluminum melting and alloying furnaces by induction systems. For manufacturing of the special aluminum alloys, ingots and deoxidants.



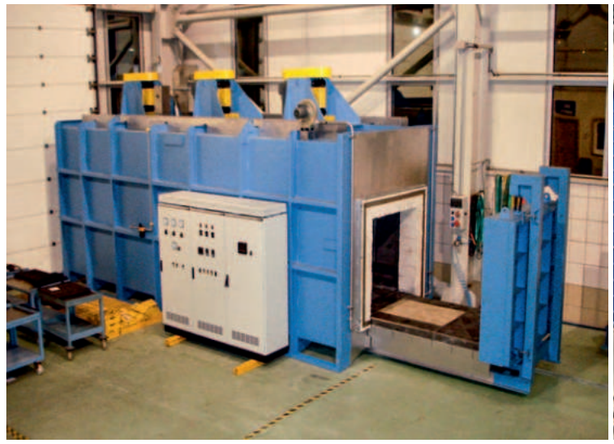












Tempering Furnaces



Bogie Heart Type Furnaces



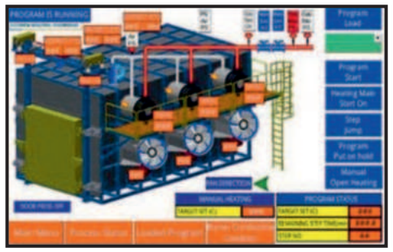
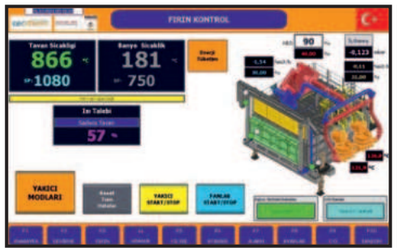
Stress Relieving Furnaces

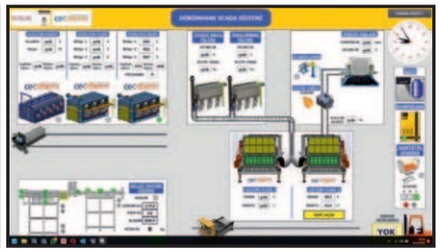
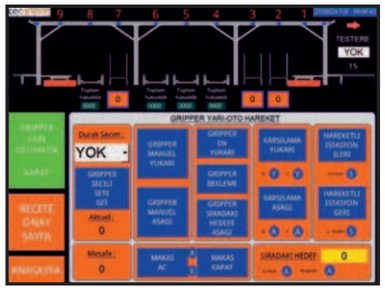


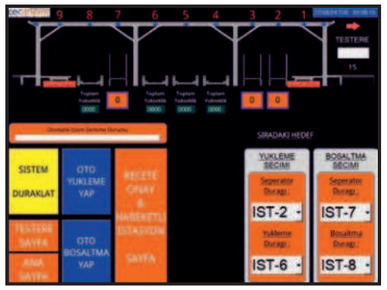
Steel Billet Re-Heating Furnaces

- Pusher Type Steel Billet Re-Heating Furnaces
- Bogie Heart Type Furnaces
- Annealing Furnaces
- Chamber Type Furnaces
- Bell Type Furnaces
- Tempering Furnaces
- Stress Relieving Furnaces
- Roller-Based Furnaces
- Walking Beam Furnaces
- Belt Conveyor Furnaces
- Continuous Heat Treatment Furnace
- Gas Atmosphere Controlled Furnaces
- Special Purpose Heat Treatment Furnaces

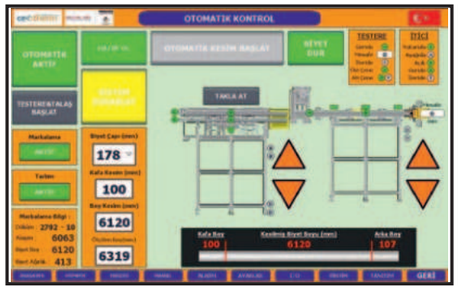
CONTROL SYSTEM

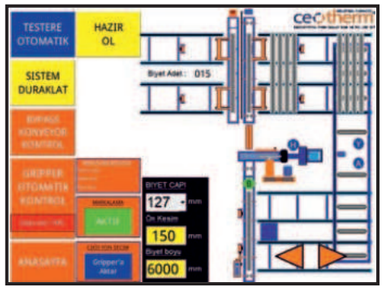













Automation and Control Systems

Our advanced automation solutions for industrial furnaces are designed to provide maximum efficiency and reliability for aluminum processing. In this context, PLC (Programmable Logic Controller), HMI (Human-Machine Interface), and SCADA (Supervisory Control and Data Acquisition) systems are effectively used to optimize the performance and control of our furnaces.

PLC (Programmable Logic Controller): The PLC systems integrated into our furnaces process data from high-precision and durable sensors, automating your processes and providing precise control. With advanced programming capabilities, you can monitor and manage every stage of your production processes. PLCs minimize system failures and ensure continuous operation of the production process.

HMI (Human-Machine Interface): Our user-friendly HMI screens provide operators with the ability to easily monitor and control the real-time status and performance of your furnaces. Graphical interfaces simplify complex processes, aiding operators in making quick and accurate decisions.

SCADA (Supervisory Control and Data Acquisition): Our SCADA system continuously monitors the performance of your furnaces with extensive data collection and analysis capabilities. This system helps optimize your processes by displaying maintenance intervals, energy consumption, and alarm statuses. SCADA's comprehensive monitoring features enhance the efficiency of your production process and enable early detection of potential issues.

Communication Protocols: Our furnaces support various communication protocols to ensure reliable system integration and data transfer. Protocols such as Modbus, Profibus, and Ethernet/IP facilitate effective communication between your furnaces and other industrial equipment, streamlining data integration.

These advanced automation technologies offer high performance and reliability for your industrial aluminum processing processes, maximizing production efficiency and making operations more manageable. Data from precise and durable sensors is processed by PLC systems to further optimize your processes and guarantee the highest quality.

OUR REFERENCES

Sistem Alüminyum
Mesan Metal
Arslan Alüminyum
Fabal
Zahit
Kontinent Gruppe
Benkam
MSC
Akpa Alüminyum A.Ş
Panda Alüminyum
Beymetal Alüminyum
Alumeta
Alubi Alüminyum
Mercan Alüminyum
Akfa
GNS Alüminyum
Karpen
Schneider Electric
Yıldız Metal
Tosun
Fonex Aluminium
Actek-MT
Tirsan
SIB
Elmetal
Gedik Döküm ve Vana
Tüp Merserize
Duduoğlu
Doğuş
Sahinler
DAL Teknik Makina
cimtas
Biser
Hakim Alüminyum
Gülermak
BASF We create chemistry
Baransan
Bor Bakır Sanayi AŞ
LBB Trade
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